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Stat 50

HW 15 #

4/10/24

Midterm #3 Sample

(1)

(a) $X \sim N\left(1.5, \frac{(0.1)^2}{200}\right)$

$$P(x > 1.520) = P\left(\frac{x - 1.5}{\frac{0.1}{\sqrt{200}}} > \frac{1.520 - 1.5}{\frac{0.1}{\sqrt{200}}}\right)$$

$$= P(Z > 2.8283) \quad = 1 - P(Z \leq 2.83)$$

$$= 1 - 0.9977 \quad = 0.0023$$

(b) $P(\bar{X} \leq b) = 0.0495$

$$-1.65 = \frac{1.5}{\sqrt{(0.1)^2/200}}$$

$$x = 1.48833$$

(b) $P(\bar{x} < 1.490) = 0.0495$

$$P\left(\frac{\bar{x} - 1.5}{0.1/\sqrt{n}} < \frac{1.490 - 1.5}{0.1/\sqrt{n}}\right) = 0.0495$$

$$\Rightarrow \frac{1.490 - 1.5}{0.1/\sqrt{n}} = -1.65$$

$$\Rightarrow \sqrt{n} = \frac{-1.65\,(0.1)}{(1.490 - 1.5)} = 16.5$$

$n = 272.25$

③ $Y \sim Exp(1/1.5)$

$P(X > 1.5\,20) = e^{-1/1.5\,(1.5\,20)}$

$= 0.3630$

② $X \sim Exp(1/12)$

(a) $P(X > 15) = e^{-(1/12)\,15}$

$= e^{-15/12} = 0.285$

(b) $P(X > 30) = e^{-(1/12)\,30}$

$= 0.08208$

③ $R_1 \sim N(100, 5^2)$

$R_2 \sim N(120, 10^2)$

$P(R_2 < R_1 + 5) = P(R_2 - R_1 < 5)$

$P(R_2 - R_1 \sim (20, 125)$

③

$$P\left(Z < \frac{S - 20}{\sqrt{125}}\right) = P(Z < -1.34)$$

$$= 0.0901$$

④

(a) $X_1 \sim \text{Poisson}(4)$

(b) $$P(X_1 = 6) = \frac{e^{-4}\, 4^6}{6!}$$

$$= 0.1042$$

(c) $T_1 \sim \text{Exp}(4)$

$$P(T_1 < 1) = 1 - e^{-4(1)}$$
$$= 0.98168$$

(e) $X_2 \sim \text{poisson}(4.3)$

$$P(X_2 = 10) = \frac{e^{-12}\, 12^{10}}{10!}$$

$$= 0.1048$$

(d) $T_2 \sim \Gamma(3, 9)$

$$P(T_2 < 2) = 1 - \sum_{j=0}^{2} \frac{e^{-36}\, 36^j}{j!}$$

$$= 1 - \frac{e^{-36}\, 36^0}{0} - \frac{e^{-36}\, 36^1}{1} - \frac{e^{-36}\, 36^2}{2}$$

d. $T_2 \sim \Gamma(3, 12)$

$$P(T_2 \leq 2) = 1 - \sum_{j}^{2} \frac{e^{-12} \, 12^{j}}{j!}$$

$$= 1 - \frac{e^{-12} \, 12^{0}}{0!} - \frac{e^{-12} \, 12^{1}}{1} - \frac{e^{-12} \, 12^{2}}{2}$$

$$= 0.9995$$

(e) $$\int_0^2 \frac{4^3 + 2 \, e^{-4t}}{\Gamma(3)} \, dt$$

(5) $\mu = 15$

$\sigma = 1.25$

$X \sim N(15, (1.25)^2)$

(a) $$P(13.5 \leq X \leq 15) = P\left(\frac{13.5 - 15}{1.25} \leq Z \leq \frac{15 - 15}{1.25} \right)$$

$$P(-1.2 \leq Z \leq 0) = P(Z \leq 0) - P(Z \leq -1.2)$$

$$= 0.5 - 0.1151 = 0.3849$$

(b) $$P(X \leq x) = 0.0281 = P\left(Z \leq \frac{x - 15}{1.25}\right)$$

$$\frac{x - 15}{1.25} = -1.91 \Rightarrow \boxed{x = 12.6125}$$

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(6) $X \sim Exp(1/2)$

(a) $P(X > 3) = 1 - P(X \leq 3) = 1 - F(3) = 1 - [1 - e^{-\frac{1}{2}3}]$

$= e^{-\frac{1}{2}(3)} = 0.223$

(b) $P(X > 7 \mid T > 4) = \dfrac{P(T > 7)}{P(T > 4)}$

$= e^{-0.5(7)} / e^{-0.5(4)}$

$= e^{-0.5(3)}$

$= e^{-1.5}$

$= 0.223$

∴ The probability that a 4-years old device last 3 additional years is same as the probability that a new device last 3 years.

(7) $X \sim U(96, 102)$

(a) $P(97 < X < 100) = \int_{97}^{100} \frac{1}{6}\, dx$

$= \frac{x}{6}\Big|_{97}^{100} = \frac{100}{6} - \frac{97}{6}$

$= \frac{3}{6} = \frac{1}{2} = .5$

(b) $Y \sim Bin(7, .5)$

$$p(x=2) = \frac{7!}{2!(5!)} \cdot .5^2 (1-.5)^5$$

$$= {}_7C_2 (.5)^2 (0.5)^5$$

$$= 0.1640625$$

Section 4.8

(1)

$X \sim U(30, 50)$

(a) $\mu = \frac{30+50}{2} = \frac{80}{2} = 40$

(b) $\sigma = \sqrt{\sigma^2} = \sqrt{\frac{(50-30)^2}{12}} = \sqrt{\frac{400}{12}}$

$\sigma = 5.7735$

(c) $P(35 < x < 42)$

$$= \int_{35}^{42} \frac{1}{40} dx = \frac{42-35}{40}$$

$$= 0.175$$

(d) $Y \sim Bin(1$

(3) $T \sim \Gamma(4, 0.5)$

(a) $\mu_T = r/\lambda = 4/.5$

$\mu_T = 8$

(b) $\sigma_T = \sqrt{\frac{r}{\lambda^2}} = \sqrt{4/(.5)^2} = 4$

(c) $P(T \le 1) = 1 - \sum_{j}^{0} \frac{e^{-0.5(4)} (0.5(4))^j}{j}$

$= 1 - \frac{e^{-0.5(4)} (0.5(4))^0}{0!} - e^{-0.5}$

$= 1 - \sum_{j=0}^{3} \frac{e^{-.5(1)} (.5)^j}{j!}$

$= 1 - \frac{e^{-.5} (.5)^0}{0!} - \frac{e^{-.5} (.5)^1}{1} - \frac{e^{-.5} (.5)^2}{2} - \frac{e^{-.5} (.5)^3}{3!}$

$= 0.00175$

(d) $P(T \ge 4) = \sum_{j=0}^{3} \frac{e^{-0.5(4)} (.5)^j}{j!}$

$$= \frac{e^{-.5(4)}(.5(4))^0}{0!} + \frac{e^{-2}(2)^1}{1} + \frac{e^{-2}(2)^2}{2} + e^{-3}\frac{(2)^3}{3.2}$$

$$= 0.8571$$

(6) $\lambda = 3.6$